Instructions:

1. This question paper contains a total of **1** page (**1** side of paper).
2. Write your name, roll number, department, and section on every side of every sheet of this booklet
3. Write final answers neatly with a blue/black pen in the given boxes.
4. **Answers written outside the box will NOT be graded.**

Total **10 Marks**

Q. 1: Write the output of the following program in the appropriate box and answer the question. (2+2+2 = 6 Marks)

```
#include<stdio.h>
void func1(int *arr){
  for(int i=1; i<=5;i++)
    (i-1)[arr] = (i%5);
}
void func2 (int a[1] , int arr){
  a[0] = arr + 3;
}
int main(){
  int arr[6] = {2,2,2,2,2};
  func1(arr+1);
  for(int i=0; i<6; i++)
        printf("%d ", i[arr] );
  printf("\n");
  func2(arr, *arr);
  for(int i=0; i<6; i++)
        printf("%d ", i[arr] );
  return 0;
}
```

Output
2 1 2 3 4 0
5 1 2 3 4 0

Will the answer change if we write func2 as:

```
void func2 (int a[1] , int arr){
  int temp = arr + 4;
  a = &temp;
}
```

Explain your answer

~~2 1 2 3 4 0~~ CompilationError

~~6 1 2 3 4 0~~

Since; 'a' is pointer to the address of first element of array. 'a' now points to value '6' of temporary variable which dies after function ends.

Q. 2: Write the output of the following sequence of instructions **long arr2[90]; int arr1[80], a1=(&arr2[60]), a2=(&arr2[10]), b1=(&arr1[60]), b2=(&arr1[10]); printf("%d %d", a1-a2 , b1-b2);** Explain your answer. (2+2 Marks)

400 200 | a1 - a2 is the difference b/w two addresses

(same with b1-b2) %d gives the difference in bytes b/w the two addresses

∴ Output (60-10) * 8 i.e. difference in no. of element * size of each element